

WILDAN ALFA RAEZEL

Junior IoT / Embedded Systems Developer

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SUMMARY

Computer Engineering undergraduate (S1 Teknik Komputer) at Universitas Amikom Yogyakarta specializing in Internet of Things (IoT) and embedded systems. Built and deployed three systems now running in production — a school attendance system, an assistive doorbell for deaf students, and an award-winning smart EV charging platform. First-place winner at ARCADIA 2026 (IoT Product category) and co-creator of a government-registered software copyright (HKI). Open to junior IoT/embedded roles, internships, and freelance work; willing to relocate.

TECHNICAL SKILLS

Microcontrollers & Boards: ESP32, ESP8266, Arduino (Uno/Mega/Nano), Raspberry Pi Pico, STM32, NodeMCU

Firmware & Languages: C/C++, MicroPython, Python, JavaScript, Dart

Protocols & Connectivity: MQTT, HTTP/REST, WebSocket, Bluetooth Low Energy (BLE), Wi-Fi, GSM/NB-IoT, LTE-M, Modbus, RS-485, Ethernet (W5500)

Sensors & Modules: extensive hands-on integration, including energy metering (PZEM-004T), RFID (RC522), environmental and motion sensors, relays, displays, RTC, SD card, and power modules (TP4056, solar panel), among many others

Hardware Prototyping: soldering, perfboard and custom PCB assembly, component-level wiring, field installation and maintenance

Backend & Cloud: Flask, Firebase Realtime Database, Firebase Firestore, Socket.IO, MySQL, PostgreSQL, Mosquitto, HiveMQ, Microsoft Azure App Service, Blynk, Adafruit IO

Mobile: Flutter, Riverpod, GoRouter, Firebase Cloud Messaging (FCM)

Tools: Git/GitHub, Docker, PlatformIO, Arduino IDE, EAGLE, EasyEDA, Fritzing, VS Code

EXPERIENCE

Laboratory Assistant — Data Acquisition Systems (IoT Concentration) 2026 – Present

Universitas Amikom Yogyakarta · Computer Laboratory

- Guide students through sensor reading, data logging, and telemetry on ESP32 in weekly practicum sessions.
- Prepare practicum materials and debug student wiring and firmware issues during lab work.

Commissioned IoT Developer (Freelance) — VIBRADoor 2025

Independent · Yogyakarta

- Built a wireless vibrating doorbell for deaf students; one bell-unit and wearable pair deployed at SLB Damayanti Sleman and in daily use since 2025.
- Owned the ESP32 firmware (Arduino framework) and hardware assembly of both devices, communicating over a self-hosted MQTT (Mosquitto) broker with a pub/sub topology that scales to multiple wristbands.
- Designed for field maintenance with a modular custom PCB and LiPo + TP4056 power; total bill of materials Rp 366K per pair.

Laboratory Assistant — Hardware Development 2025

Universitas Amikom Yogyakarta · Computer Laboratory

- Assisted the practicum on microcontroller fundamentals and component-level wiring with Arduino and ESP32; maintained lab equipment inventory and prepared weekly kits.

PROJECTS

Presensi SMPN 2 Paliyan — IoT School Attendance System 2026 – Present

Team project with SMPN 2 Paliyan as partner school · presensi-smpn2-paliyan.my.id

- RFID + keypad attendance device installed at the school gate and used for daily attendance, with a real-time admin dashboard and a public parent portal.
- Developed the ESP32 firmware, assembled the gate hardware (RC522 RFID, keypad, LCD, DS3231 RTC, W5500 Ethernet, SD card, UPS), and built the Flask + Firebase Firestore backend serving as the single data gateway.
- Engineered fault tolerance: SD-card store-and-forward through network outages, UPS backup through power cuts, and watchdog + scheduled reboots for unattended 24/7 operation.

PAYCAY — Smart EV Charging Station (IoT + Web)

2025 – 2026

Solo build · github.com/alfa219/paycay-

- Won 1st place, IoT Product category, at ARCADIA 2026 (Faculty of Computer Science, Universitas Amikom Yogyakarta).
- Built end-to-end: ESP32 firmware (PZEM-004T energy metering, RFID tap-to-start, AC relay control), Flask backend with a real-time Socket.IO dashboard and Firebase Realtime Database; deployed live on Azure App Service with GitHub Actions CI/CD.
- Implemented automatic balance deduction from kWh readings and a complete REST API covering the full charging-session lifecycle.

PAYCAY Mobile — Flutter Companion App

2025 – 2026

Solo build · github.com/alfa219/app-paycay

- Dual-role (user/admin) app with real-time station monitoring via Firebase RTDB, a digital wallet using atomic Firestore transactions, and session recovery that survives the app being killed mid-charge.

EDUCATION

Universitas Amikom Yogyakarta — S1 Teknik Komputer (Computer Engineering)

2023 – Present

Concentration: Internet of Things · Expected graduation 2027

CERTIFICATIONS

- CCNA: Enterprise Networking, Security & Automation — Cisco Networking Academy, Jul 2025 (ID: 59fcddb5-bfce-44ef-8d7d-b431c87be388)
- CCNA: Switching, Routing & Wireless Essentials — Cisco Networking Academy, Mar 2025 (ID: f8e80bd1-e91e-4cf0-919f-0840ed7bcd1)
- CCNAv7: Introduction to Networks — Cisco Networking Academy, Aug 2024 (ID: 6bd505b3-7acc-4f70-b5ba-05424ae229f1)

AWARDS & INTELLECTUAL PROPERTY

- 1st Place, IoT Product — ARCADIA 2026, Universitas Amikom Yogyakarta (winning entry: PAYCAY), Feb 2026.
- Honorable Mention (Juara Harapan II), IoT category — Indonesia National IT Competition (INDONERIS) 2025, Universitas Amikom Purwokerto, Dec 2025.
- Registered Software Copyright (Hak Cipta / HKI) — "Emergency Input & Alert Station (EIAS) v1.0", listed co-creator; DJKI reg. no. 001273262, 2026.